Favorite thing about this week?

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Q1

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340

CS 340

Python

Updates

1. MP6 Wallet due tonight

2. MP7 out today (this afternoon)

3. Exam-2 2nd chance on Thursday

Python

Today's LGs - Be able to read and write python code

Be able to complete MP7 in python

What is your experience with Python?

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Q2

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Agenda

1. Python Intro (cheat sheet)
2. Python Vs. C
3. Python Sticking Points
4. MP7
5. MP6

What prints?

```
x = 4
y = [9, 2, 5]
y.append(x)
if 4 in y:
    print("yay")
else:
    print("no")
```

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Q3

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A few favorites about Python

```python
print('hello')
x = {'abc':10, 'bye':6}
print(x)

y = [1, 4, 90]
if 10 in y:
    print('hi')
print('bye')
```

PROBLEMS OUTPUT TERMINAL 2

TERMINAL bash

```
root@0e9b25a8fb50:/workspaces/mp7-testing# python3 test_class.py
hello
{'abc': 10, 'bye': 6}
```

Python Cheat Sheet

1. online
2. in person

"hi there person"

.split() = ["hi", "there", ...

x = [1, 2, 3]

x[-2::]

Python Cheat Sheet

```
for i in range(0,10):
    ...
```

0 ... 9

```
len([1,2])
```

2

Python Cheat Sheet

Python Cheat Sheet

Challenge: Does this code pass the assert?

```
def func(nums, k):
    result = []
    for i in range(len(nums) - k + 1):
        window = nums[i:i+k]
        result.append(max(window))
    return result

assert func([1, 3, 2, 5, 4], 3) == [3, 5, 5]
```

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yes → no issue

no → raises assertionError

Python Vs. C , C++

C

1. compile to machine code
2. machine code is run on CPU

Python

- compile into simpiler python
- loaded by the python interpreter
- executed on CPU

C program

errors
performance
types

Errors

```
1  print('hello')
2  x = y
```

PROBLEMS 1 OUTPUT TERMINAL 2

TERMINAL

```
root@0e9b25a8fb50:/workspaces/mp7-testing# pytho
hello
Traceback (most recent call last):
  File "/workspaces/mp7-testing/test_class.py",
    x = y

NameError: name 'y' is not defined
root@0e9b25a8fb50:/workspaces/mp7-testing#
```

Performance Favors Library Code

```
total = 0
for x in range(10000000):
    total += x
print(total)
```

slow

```
total = sum(range(10000000))
print(total)
```

fast

Dynamically Typed

```
x = 'hi'
x = 5
x = [4, 10, "hi"]
```

```
def func(y):
    y = y+y
    return y

x = [4, "hi"]
x = func(x)
x = func(5)
```

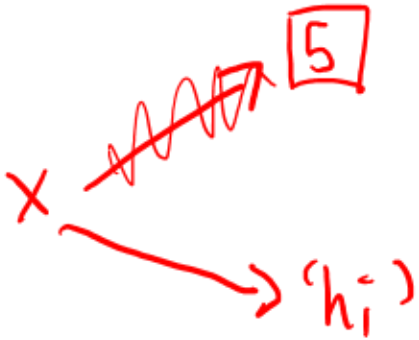
list

int

C++ / C

int x = 5;

Python



Tricky Examples

```
x = 5
y = x
y += 5
print(x)
```

y = y + 5

5

x → 5

y → 10

```
x = [5, 6, 7]
y = x
y.append(8)
print(x)
```

x → [5, 6, 7, 8]

y → [5, 6, 7, 8]

in-place

[5, 6, 7, 8]

What prints?

```
z = [1, 2]
y = [1, 2]
y.append(6)
print(z)
```

z → [1, 2]

y → [1, 2, 6]

[1, 2]

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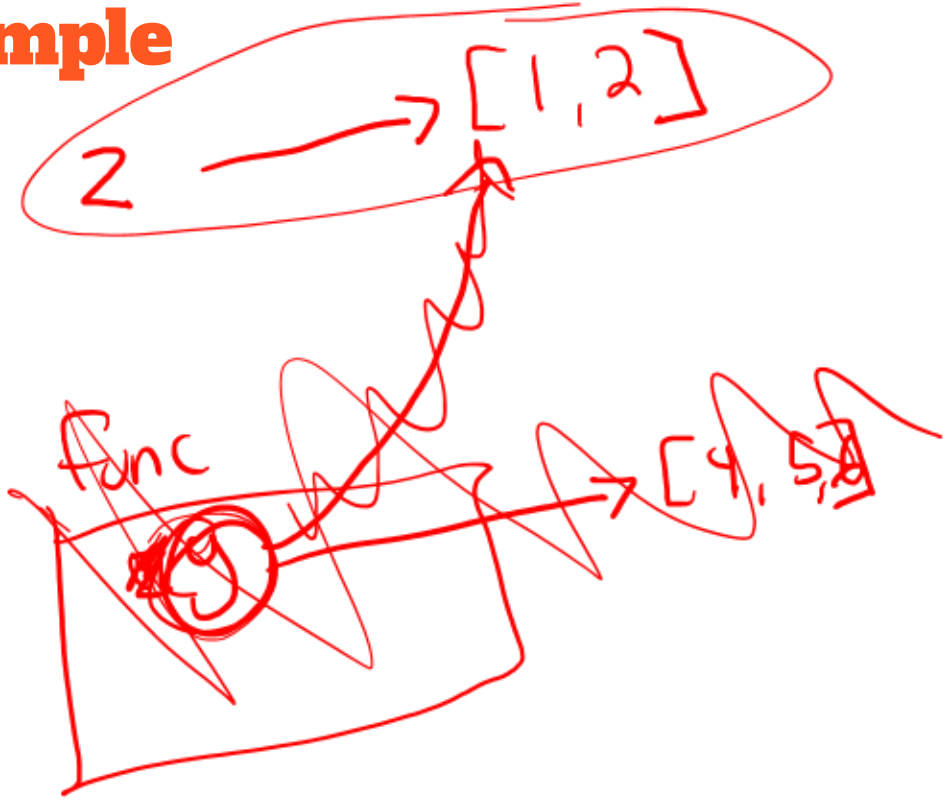
Tricky Function Example

```
def func(y):
    y = [4, 5]
    y.append(6)

z = [1, 2]
func(z)
print(z)
```

[1, 2]



What prints?

```
def func(y):
    y = []
    y.append(6)

z = [1, 2]
func(z)
print(z)
```

z → [1, 2, 6]

func

y

[1, 2, 6]

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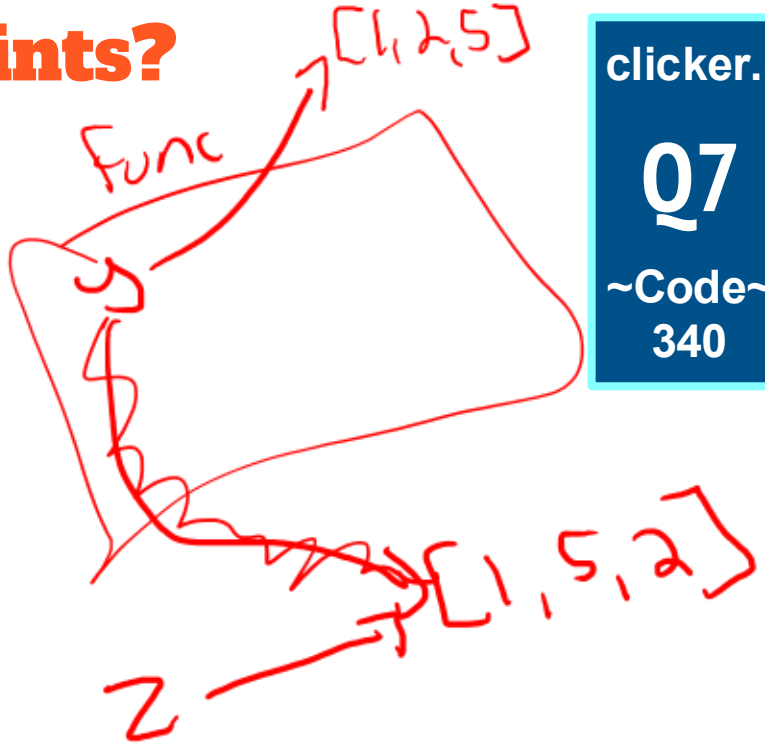
Challenge: What prints?

```
def func(y):
    y=sorted(y)
    ? [1,2,5]

z = [1, 5, 2]
func(z)
print(z)
```

[1, 5, 2]



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Optional Types for Readability

```python
def func(y):
    y += y
    return y

x = 5
x = func("x")
print(x)
```

```python
def func(y: int) -> int:
    y += y
    return y

x: int = 5
x = func(x)
print(x)
```

Scope

- only
  - function
  - module (file)

```
for i in range(0,5):
    x = 5
print(x)
if 6 == 5:
    x = 3
print(x)
```

```
1   x = 5
2   def func(y: int):
3       w = 6
4       global x
5       x += y
6   func(2)
7   print(x)
```

print(x)

~~print(w)~~

What prints?

```
x = 5
for i in range(0, 3):
    x = i
print(x)
```

i will be 0, 1, 2

a) 5

b) 1

c) 2

d) 3

e) error

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Python Sequences

```python
def func():
    return 10, "hi", {3: "hi", 4:"bye"}

a, b, c = func()
print(c)
```

a → 10
b → "hi"
c → {

PROBLEMS OUTPUT TERMINAL 2

TERMINAL

```
root@0e9b25a8fb50:/workspaces/mp7-testing# python3 test_class.py
{3: 'hi', 4: 'bye'}
```

MP7 - Scheduler Tests

Python or MP7 Questions?

MP6 Help